Prompt Then Refine: Prompt-Free SAM-Enhanced Collaborative Learning Network for Detecting Salient Objects in Underwater Images

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***Abstract*—RGB–depth underwater salient object detection (USOD) poses considerable challenges, such as uneven lighting, visual interference, and image blur, which limit the effectiveness of traditional approaches. The segment anything model (SAM), known for its robust segmentation capabilities, offers a promising alternative. However, SAM depends on prompt labels (e.g., points, boxes, and masks) to perform effective resources typically unavailable in USOD datasets. To address this, we propose SAM-CLNet, a prompt-free, SAM-enhanced collaborative learning network comprising three main components: 1) SAM; 2) a mask prompt generator (MPG); and 3) a region-aware attention collaborative learning loss (RCL). In our framework, pseudo-mask prompts generated by MPG were used as input prompts for SAM, helping to offset performance degradation due to the absence of manual labels. Simultaneously, RCL leveraged high-quality SAM predictions to refine MPG, enhancing its feature extraction while minimizing the impact of low-quality pseudo-prompts on SAM. This cyclic feedback mechanism facilitated mutual improvement in detection accuracy. In addition, we introduced a U-Adapter module to adapt SAM for underwater imagery and incorporated a frequency cross-attention fusion module in MPG to integrate RGB and depth information. The region-aware attention in RCL further targeted challenging regions by comparing SAM's predictions with MPG's pseudo-mask. Experiments on the USOD10K and USOD datasets demonstrated that SAM-CLNet outperformed existing methods and generalized effectively across five public salient object detection (SOD) benchmarks. Code is available at https://github.com/BeibeiIsFreshman/SAM-USOD**

***Index Terms*—Collaborative learning, salient object detection (SOD), segment anything model (SAM), transfer learning, underwater images.**

I. Introduction

UNDERWATER salient object detection (USOD) involves identifying and localizing the most prominent objects in underwater images and plays a critical role in underwater robotic vision and the autonomous navigation of submersible vehicles [1], [2]. However, underwater imaging introduces inherent challenges such as uneven lighting, visual interference, and image blurring, which hinder object recognition. To advance research in USOD, Islam et al. [3] constructed the first USOD dataset; however, its limited scale and lack of environmental realism restricted its practical applicability. To address these limitations, Hong et al. [4] introduced USOD10K—the first RGB–depth USOD dataset—along with a strong baseline model, TC-USOD, which demonstrated that RGB–depth inputs outperformed RGB-only inputs. Building on this foundation, Jin et al. [5] proposed a self-paced learning and depth-emphasis network that accounted for varying sample difficulty, further improving detection performance. More recently, Ma et al. [6] developed TJUP-USOD, the first RGB-polarization USOD dataset. However, because TJUP-USOD was not collected in underwater environments, it lacked environmental authenticity. Empirical results continue to show that RGB–depth USOD10K outperforms RGB-polarization datasets. Therefore, this study focuses on the RGB–depth USOD10K and USOD datasets.

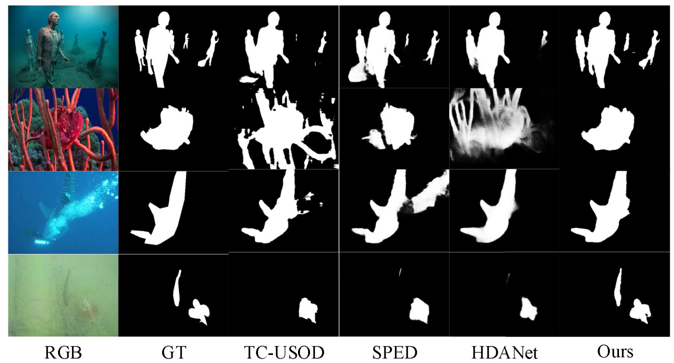

Fig. 1. RGB image, GT image, TC-USOD [4], SPED [5], HDANet [7], and our (SAM-CLNet) predicted image.

Current state-of-the-art (SOTA) methods on the USOD10K dataset include the robust benchmark network TC-USOD [4], the self-paced learning mechanism combined with depth emphasis networks [5], and the enhancing network with physics-inspired multimodal joint learning [7]. However, as Fig. 1 shows, in complex underwater environments, all three

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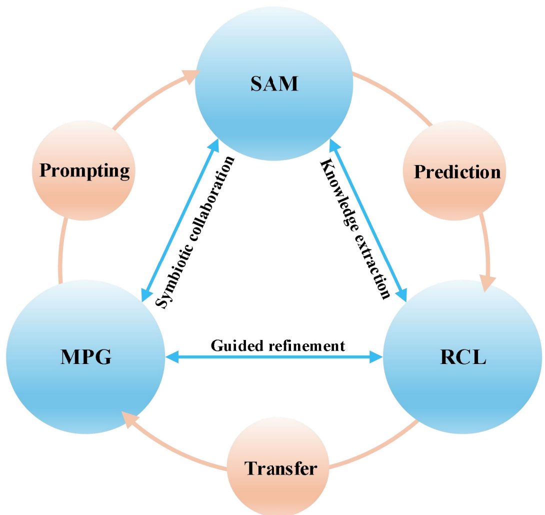

Fig. 2. Schematic of SAM-CLNet. MPG and SAM engage in symbiotic collaboration: MPG prompts SAM with pseudo-masks, SAM provides predictions to RCL for knowledge extraction, and RCL guides MPG refinement through knowledge transfer. This cyclic mechanism enables continuous mutual improvement.

models exhibit limited recognition accuracy and insufficient adaptability to environmental variations. Therefore, how to improve the performance of USOD systems remains a critical research question that needs to be addressed.

The segment anything model (SAM) [8], introduced by Meta AI as a foundation vision model, offers a promising direction for USOD. Trained on over 1 billion samples, SAM demonstrates strong generalizability and is theoretically well-suited for general salient object detection (SOD) tasks [9]. However, its high performance depends on precise prompt label input, such as points, boxes, or masks, which are unavailable in the largest USOD dataset, USOD10K. Moreover, manually generating these labels is both labor-intensive and time-consuming [10], [11]. Recent studies have explored two main strategies to adapt SAM for downstream tasks. The first involves integrating adapter modules into SAM's image encoder to improve task-specific adaptability while preserving its core architecture. However, this approach still relies on prompt labels to perform effectively [12]. The second strategy retains only SAM's image encoder for RGB feature extraction, discarding the prompt encoder and related components. While this method has shown success in certain applications, it necessitates extensive architectural redesign and fine-tuning to suit specific tasks [13], [14], [15], [16], [17].

We aim to apply SAM to USOD without relying on prompt labels or altering the original model architecture. To achieve this, we propose a prompt-free, SAM-enhanced collaborative learning network (SAM-CLNet) (see Fig. 2). SAM-CLNet operates through a cyclical framework composed of three main components: 1) SAM; 2) a mask prompt generator (MPG); and 3) region-aware attention collaborative learning loss (RCL). First, MPG generates a pseudo-mask prompt that serves as input to SAM's prompt encoder. Based on this input, SAM produces high-quality prediction maps. Then, these predictions are passed back to MPG through RCL, enabling MPG to extract more representative features. Simultaneously, SAM refines MPG while remaining unaffected by the low-quality pseudo-mask, thereby preserving its own performance [18]. This cyclic feedback mechanism iteratively enhances the overall detection accuracy of SAM-CLNet. The architectures of the three core components—SAM, MPG, and RCL—are outlined below. A U-Adapter is incorporated into SAM's image encoder to improve its adaptability to underwater imagery, and all redundant sparse prompt components are removed. MPG adopts a U-Net structure and integrates a frequency cross-attention fusion module to combine RGB and depth features extracted via a transformer backbone. Finally, RCL introduces region-aware attention to reconcile discrepancies between SAM's predictions and MPG's pseudo-mask in visually challenging regions.

We begin by adopting SAM as the base network and integrating a U-Adapter module into its image encoder to accommodate the distinct characteristics of underwater imagery while preserving SAM's original performance. The U-Adapter consists of a prompt generator and a prompt tuning module, both of which enhance adaptability to underwater scenes through dynamic prompt generation and feature-level fusion. Next, the proposed MPG generates a pseudo-mask that serves as input to SAM's prompt encoder. MPG adopts a U-Net architecture and integrates a transformer backbone to jointly process RGB and depth inputs for pseudo-mask generation. At its core, MPG incorporates a frequency cross-attention fusion module, which fuses multiscale features and extracts complementary information from RGB and depth modalities using a combination of frequency-domain attention, cross-modal attention, wavelet decomposition, and an adaptive gating mechanism. Finally, the proposed RCL is introduced as a loss term that prioritizes regions where SAM's predictions and MPG's outputs diverge, assigning greater weight to these challenging regions during training. As Fig. 2 shows, this collaborative learning loss enables the transfer of high-quality predictive knowledge from SAM to MPG, refining the pseudo-mask prompts to better approximate SAM's ideal output. This mechanism also protects SAM from degradation due to a noisy pseudo-mask. As MPG improves, the quality of its generated masks increases, which, in turn, enhances SAM's predictions—establishing a mutually reinforcing feedback loop that progressively improves overall detection performance.

Our main contributions are as follows.

1) We propose SAM-CLNet, a prompt-free, SAM-enhanced network that addresses performance degradation in SAM caused by the absence of prompt labels. The effectiveness of SAM-CLNet for USOD is validated on the USOD10K and USOD datasets, and its generalization capability is further demonstrated across five additional SOD benchmarks.

2) We introduce a U-Adapter module that improves SAM's adaptability to underwater imagery by dynamically generating prompts and performing layer-wise feature fusion within the image encoder. This design enables effective adaptation to USOD tasks while preserving SAM's original segmentation performance.

3) We develop an MPG module that integrates RGB and depth features through a frequency cross-attention fusion unit, combining frequency-domain attention, cross-modal attention, wavelet decomposition, and adaptive gating to extract complementary information across modalities.

4) We propose RCL, a region-aware collaborative learning loss that serves as the central optimization mechanism in SAM-CLNet. RCL facilitates cyclic learning between SAM and MPG by automatically identifying challenging regions and emphasizing these during training.

A review of related work is presented in Section II. Section III details the architecture of SAM-CLNet and its three core components. Section IV reports the experimental results on the USOD10K and USOD datasets, evaluates generalization across five additional benchmarks, and provides an analysis of representative failure cases. Section V concludes this article and outlines future research directions.

II. Related Work

A. RGB–Depth USOD

RGB–depth SOD improves detection accuracy by fusing RGB texture information with spatial cues from depth maps [19]. As a specialized subdomain, USOD has received growing attention in the computer vision community, aiming to detect salient regions in underwater imagery where visual conditions are often degraded.

In terrestrial environments, various RGB–depth SOD methods have been proposed. For instance, Fan et al. [20] introduced a bifurcated backbone architecture that fused RGB and depth features, establishing a foundation for subsequent multimodal approaches. Zheng et al. [21] proposed a triplet transformer embedding module that enhanced feature representation through coordinated integration of multiple submodules. Li et al. [22] addressed the problem of low-quality depth maps by designing a network based on hierarchical alternate interactions, providing the first dedicated solution to this common limitation in RGB–depth SOD. Zhang et al. [23] proposed a cross-modality discrepant interaction network in which features from different layers were treated separately and fused using distinct modules. Cong et al. [24] introduced a point-aware interaction and CNN-induced refinement network that improved long-range dependency capture compared with conventional CNNs. Wei et al. [25] developed an edge feature enhancement and global information attention network that restored salient object details by refining edge information. Wu et al. [26] presented a source-free depth-guided object pop-out network that transferred depth information into semantic representations. Sun et al. [27] designed a cascaded and aggregated transformer framework to minimize information loss and redundancy during multiscale feature fusion. Zhong et al. [28] proposed a multiscale awareness and global fusion network that achieved strong performance by applying differentiated fusion strategies to various feature types. Recently, Gao et al. [29] introduced an efficient lightweight network aimed at reducing model parameters and computational overhead.

Although these RGB–depth SOD models have shown strong performance in terrestrial scenes, they encounter major limitations when applied to underwater environments due to unique visual degradations such as scattering, turbidity, and low contrast. To advance USOD development, Islam et al. [3] pioneered the first USOD dataset; however, the proposed dataset had a limited scale and did not attract widespread attention to this field. It was not until Hong et al. [4] introduced the first large-scale USOD dataset, USOD10K, that the field garnered more attention. Along with the dataset, Hong et al. [4] proposed a robust benchmark network, TC-USOD. Jin et al. [5] improved detection performance by proposing a self-paced learning mechanism combined with depth-emphasis networks. Subsequently, Liu et al. [7] proposed an enhanced network with physics-inspired multimodal joint learning that redefined the field's development through dataset augmentation and visual enhancement. Furthermore, Ma et al. [6] developed TJUP-USOD, the first RGB-polarization USOD dataset. However, the incorporation of the polarization modality did not provide significant performance improvements compared to the addition of the depth modality. Consequently, most researchers continue to focus on USOD10K as the primary dataset for USOD research.

B. Segment Anything Model

Recent advances in computer vision have been shaped by the release of the SAM by Meta AI, which has driven notable progress in semantic segmentation tasks [8]. Since its introduction, researchers have explored various strategies to fine-tune SAM, achieving promising results across diverse domains. For example, Chen et al. [12] proposed SAM-Adapter, which introduced simple yet effective adapters that embed domain-specific information or visual prompts into the segmentation network, laying the groundwork for further fine-tuning. Gao et al. [13] presented a multiscale, detail-enhanced version of SAM by removing the prompt encoder and redesigning the decoder to reduce its reliance on precise object prompts. Xiong et al. [14] introduced EfficientSAM, which, for the first time, applied knowledge distillation to transfer learned representations from SAM to a lightweight model. Cheng et al. [15] developed H-SAM, a prompt-free variant of SAM tailored for medical imaging using a two-stage hierarchical decoding scheme. Zhang et al. [16] proposed GoodSAM, which integrated semantic information via a teaching assistant mechanism to generate fused representations and facilitate knowledge transfer. Liu et al. [17] presented a network ensemble based on SAM to support multimodal inputs, addressing the model's original limitation to RGB-only data. In addition, recent studies have extended SAM to a wide range of downstream tasks [9], [30], [31].

Building on these developments, we propose SAM-CLNet, which incorporates an auxiliary network (MPG) to generate mask prompts as inputs to SAM and integrates a U-Adapter into SAM's image encoder. This architecture enables SAM to learn underwater-specific features while preserving its original parameters, thereby balancing task-specific adaptation with model integrity.

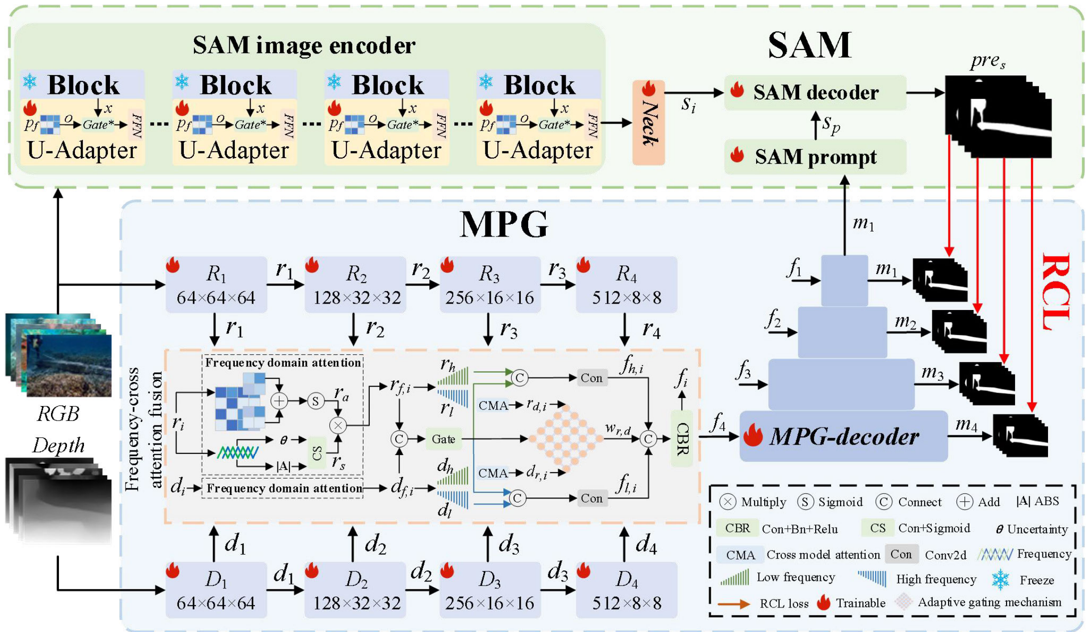

Fig. 3. SAM-CLNet diagram, in which the SAM encoder is shown with a light-green background, the MPG encoding part is in a light-blue background, and the red arrows indicate RCL. The overall process of the model is as follows: first, the RGB and Depth images are input into the MPG to generate the pseudo-mask m_1. Then, the RGB image and generated pseudo-mask m_1 are separately input into the SAM image and prompt encoders to generate the final prediction result, pre_s. Finally, through the RCL, the knowledge in pre_s is transferred back to the MPG, further refining the pseudo-mask, forming a loop that improves the overall model prediction performance.

C. Collaborative Learning

Collaborative learning refers to the process in which multiple learners interact to improve overall learning effectiveness [32], [33], [34]. Wang et al. [35] proposed a two-stage deep collaborative learning module that replaces traditional convolutional layers, enhancing visual recognition by enabling feature-level interaction. To our knowledge, this was the earliest effort to incorporate collaborative learning into deep neural networks, though it initially attracted limited attention. With the evolution of online knowledge distillation techniques, collaborative learning has since emerged as an active and distinct research area [36]. Wu et al. [37] introduced a CNN-CNN-transformer rectified collaborative learning framework that enables cyclic knowledge transfer between CNNs and transformers, improving performance in medical image segmentation. Liu et al. [38] developed a collaborative adaptive metric distillation strategy that aggregates knowledge across learners through a cooperative scheme. Xu et al. [39] proposed a dual-classifier collaborative learning framework in which two classifiers iteratively refine each other's predictions via mutual learning. Chen et al. [40] presented a style-adaptive pooling network guided by two pretrained models, forming a collaborative learning setup to accelerate training of a new network. Aksoy et al. [41] designed a multilabel noise-robust collaborative learning method to mitigate the effects of label noise in CNN training. Recently, Guo et al. [42] proposed a fast multiview subspace clustering algorithm based on a local-global anchoring mechanism, where local and global anchors were collaboratively learned to enhance clustering accuracy.

Building on prior studies, we propose SAM-CLNet, a collaborative framework that establishes cyclic knowledge transfer between SAM and MPG. In this design, MPG generates pseudo-masks that are fed into SAM's prompt encoder, while high-quality predictions from SAM are transferred back to MPG through a novel RCL mechanism. This feedback process refines MPG to generate masks that more closely approximate SAM's ideal output. As MPG improves, the quality of its generated masks increases, which in turn enhances SAM's predictions—creating a virtuous feedback loop that progressively improves overall system performance.

III. Proposed SAM-CLNet

A. Overview

As Fig. 3 shows, the proposed SAM-CLNet consists of three core components: SAM, MPG, and RCL. MPG produces a pseudo-mask, denoted as m_1, which was used as input to SAM's prompt encoder. Based on this input, SAM generated a high-quality prediction map pre_s, which was then used to guide the refinement of subsequent masks $[m_1, m_2, m_3, m_4]$ generated by MPG through the proposed RCL loss. This process encouraged the generated pseudo-mask to better approximate SAM's ideal output. As the MPG module improved, the quality of its output correspondingly increased. These higher-quality masks further enhanced SAM's predictions, forming a virtuous feedback loop that continuously strengthened the performance of the entire SAM-CLNet framework. Sections III-B through III-C describe the SAM, MPG, and RCL components in detail. During training, both pre_s and m_i were supervised by the ground truth (GT) maps.

B. U-Adapter: Specialized USOD Adapter for SAM

The upper portion of Fig. 3 illustrates the original SAM architecture, which included three components: an image encoder, a prompt encoder, and a mask decoder. RGB images were first processed by the image encoder to produce encoded features, denoted as s_i. Concurrently, the pseudo-mask generated by MPG was passed to the prompt encoder to produce s_p. Both s_i and s_p were then forwarded to the mask decoder to generate the final prediction map, pre_s. We enhanced SAM in two keyways. First, we integrated a custom-designed U-Adapter into SAM's image encoder to improve domain adaptation for USOD tasks. This component was designed to enhance the encoder's ability to extract discriminative features from underwater imagery. Second, we removed sparse prompt components from the original prompt encoder and mask decoder to ensure architectural compatibility with SAM-CLNet. As this modification did not represent a core innovation of our work, its implementation details were not elaborated. In Fig. 3, modules marked with snowflakes were excluded from parameter updates during training.

1) U-Adapter: Prior studies have explored full fine-tuning of SAM for SOD; however, this approach often resulted in excessive trainable parameters and might introduce performance degradation [13]. A more efficient alternative involves transferring a limited set of trainable parameters into SAM, thereby enabling adaptation to downstream tasks while preserving the original model's performance [11], [12], [42], [43]. In our preliminary experiments, we observed that existing SAM adapters did not perform effectively in the context of USOD. To address this, we proposed the U-Adapter, a lightweight and task-specific module designed to adapt SAM for USOD scenarios. The U-Adapter consisted of two submodules: a prompt generation module and a prompt tuning module.

The prompt generation module produced prompt vectors that modulated the behavior of transformer blocks within SAM's image encoder. It generated these vectors based on the contextual features of the input, enabling customized feature representations for each image. Specifically, using truncated normal distribution initialization [45], a set of learnable base prompt vectors p_b was initialized for each encoder layer. Subsequently, we employed an independent *LayerAdapter* for each layer's prompt vectors p_b to generate intermediate prompts p_a with different styles. Finally, the intermediate vectors were refined through *FeatureAdapter* weighting based on the global context information of input features F_b. Specifically, the *FeatureAdapter* generated attention weights for each prompt vector through SoftMax normalization, and applied these weights to p_a to produce the final adaptive prompt vectors p_f. This three-stage mechanism enabled the prompts to dynamically adapt to both layer-specific representations and the characteristics of individual input images, thereby enhancing the model's adaptability across diverse visual inputs. This process was formulated as

$$p_b = Rand\left(L \times N_p \times D\right) \tag{1}$$

$$p_a = LayerAdapter\left(p_b\right) \tag{2}$$

$$p_f = \frac{1}{B}\sum_{b=1}^{B}\left(p_a \times Soft\left(FeatureAdapter\left(F_b\right)\right)\right) \tag{3}$$

where *Rand*() denotes a random generation operation, L the number of block layers, N_p the number of prompt vectors per layer, and D the feature dimension, B the batch size, *Soft* the SoftMax operation, and F_b the feature context in batch B (we used the spatial average pooling in this study), each adapter consisted of a two-layer MLP with LayerNorm and GELU activation to capture layer-specific stylistic variations.

The prompt tuning module fused the generated prompt vectors with input features using an attention mechanism to highlight informative regions. It modulated the influence of the prompts via a gating function. First, both the feature input x and the generated prompt vectors p_f were projected into query, key, and value spaces. Then, a multihead attention mechanism was applied to these projections to produce the intermediate attention output o_a

$$q = Linear\left(x\right) \tag{4}$$

$$k, v = Linear\left(p_f\right) \tag{5}$$

$$o_a = MultHeadAtt\left(q, k, v\right). \tag{6}$$

Here, *MultHeadAtt*() denotes the multihead attention mechanism.

The gating mechanism regulated the degree to which each feature position in x incorporated prompt-based information to generate the final output x_z. When a feature aligned strongly with a prompt vector, the corresponding gate value approached one, enabling full influence. Conversely, when the feature was weakly related, the gate approached zero, suppressing the prompt's effect. This soft-switch behavior enabled adaptive modulation of external guidance. It was defined as

$$g = Gate\left(Cat\left(x, o_a\right)\right) \tag{7}$$

$$x_z = x + g \times o_a. \tag{8}$$

Here, *Gate*() denotes the gating operation, and *Cat*() represents the concatenation function.

Finally, the gated feature x_z was passed through a residual feedforward network to improve its representational capacity, producing the final output x_o

$$x_o = x_z + FFN\left(LayerNorm\left(x_z\right)\right) \tag{9}$$

where *FFN*() denotes a feedforward network, and *LayerNorm*() refers to the layer normalization operation.

The U-Adapter was an adaptive prompt-tuning architecture designed to apply customized enhancements to each transformer block within SAM's image encoder. It identified the network depth using layer indices and generated distinct prompts tailored to varying levels of semantic abstraction. These prompts were dynamically modulated based on both positional and contextual cues from the input data. They were integrated with the encoder's original features via a combination of multihead attention and gating, which ensured effective enhancement of both shallow local patterns and deeper semantic structures. This design allowed SAM to adapt to underwater imagery without requiring full fine-tuning, thereby preserving its original segmentation capabilities while improving performance in domain-specific tasks.

C. Mask Prompt Generator

The lower portion of Fig. 3 illustrates the architecture of the MPG, which comprises a PVTv2 [46] backbone, a frequency cross-attention fusion module, and a decoder. The decoder employed a conventional design based on upsampling and convolution for progressive layer-wise reconstruction and was not discussed further in this article. RGB and depth images were first passed through the pvt-v2 backbone to extract multiscale features $\{r_1, r_2, r_3, r_4\}$ and $\{d_1, d_2, d_3, d_4\}$, respectively. Then, these features were input into the frequency cross-attention fusion module to produce fused representations $\{f_1, f_2, f_3, f_4\}$, which were subsequently decoded to generate four corresponding masks $\{m_1, m_2, m_3, m_4\}$. Among these, the mask m_1 served as the pseudo-prompt input for SAM's prompt encoder.

1) Frequency Cross-Attention Fusion Module: Multiscale RGB–depth features contain rich complementary information that must be fused effectively to ensure optimal detection performance. Simple fusion methods such as element-wise addition or direct concatenation fail to capture complex cross-modal dependencies, often leading to feature degradation or information loss. Existing methods often rely on single-dimensional attention or hierarchical fusion strategies, which either underuse the complementary nature of RGB and depth features or increase computational cost [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [47]. Many also overlook the potential of frequency-domain information. To address these issues, we propose a frequency cross-attention fusion module that integrates RGB and depth information using frequency-domain attention, cross-modal attention, wavelet decomposition, and adaptive gating, achieving multiscale fusion while preserving structural and semantic details.

First, the features r_i and d_i were processed through a convolutional block attention module (including pooling, fully connected layers, and a sigmoid function) to obtain r_a and d_a, which weight each channel to enhance important features and suppress less relevant ones. These features were then passed through a frequency-domain attention mechanism, producing r_s and d_s, which combine spatial attention with frequency-based analysis to extract key visual cues. Finally, r_a and d_a (representing channel importance) were fused with r_s and d_s (representing visual cues) to produce the final frequency-enhanced features, $r_{f,i}$ and $d_{f,i}$, using a 2-D Fourier transform to highlight salient structural patterns. A similar process was applied to both depth and RGB features, as shown in Fig. 3, so the details for depth features were omitted. The entire process was expressed as

$$r_a\left(d_a\right) = Sig\left(FC\left(AP\left(x\right)\right) + FC\left(MP\left(x\right)\right)\right) \tag{10}$$

$$mag = \frac{\left|FFT2D\left(x\right)\right|}{\max\left(\left|FFT2D\left(x\right)\right|\right) + \varepsilon} \tag{11}$$

$$pha = \angle FFT2D\left(x\right) \tag{12}$$

$$r_s\left(d_s\right) = Sig\left(Con\left(Cat\left(MeanC\left(mag\right), MeanC\left(pha\right)\right)\right)\right) \tag{13}$$

$$r_{f,i}\left(d_{f,i}\right) = x \times r_a\left(d_a\right) + x \times r_s\left(d_s\right) \tag{14}$$

where x represents either r_i or d_i, FC denotes a fully connected layer, $AP()$ and $MP()$ represent adaptive average pooling and max pooling, respectively, $FFT2D()$ is the 2-D fast Fourier transform, $||$ indicates magnitude computation (i.e., amplitude of the frequency spectrum), $\angle$ denotes phase calculation, $Sig()$ is the sigmoid activation function, $Con()$ represents convolution, $MeanC()$ indicates channel-wise averaging, and ε is a small constant added for numerical stability to prevent division by zero.

Next, cross-modal attention was applied to enable bidirectional information exchange between the RGB and depth modalities (ri and di, respectively). Given the inherent asymmetry of these features, in which RGB provides rich textural and semantic cues while depth captures the geometric and spatial structure, cross-modal attention was beneficial [48], [49]. Previous studies have shown that these asymmetric features can enhance each other as RGB helps refine depth boundaries in textureless areas, while depth provides geometric constraints to improve RGB-based recognition in challenging lighting conditions [50], [51]. This interaction creates enriched feature maps ($r_{d,i}$ and $d_{r,i}$), with each modality enhanced by context from the other. Unlike symmetric fusion methods that assume equal contribution from both modalities, our cross-modal attention mechanism adaptively weighs each modality based on local feature reliability [52], optimizing their complementary strengths and mitigating potential negative transfer from low-quality features. This process enables mutual reinforcement and addresses the common issue of limited information flow in traditional fusion strategies, ultimately improving feature representation robustness and expressiveness. The attention formulation was defined as follows:

$$q_f, k_b, v_b = reshape\left(W_{qkv}\left(x_f, x_b\right)\right) \tag{15}$$

$$a = soft\left(q_f \times k_b\right),\ o = v_b \times a^T \tag{16}$$

$$r_{d,i}\left(d_{r,i}\right) = \alpha \times o + x_f \tag{17}$$

where q_f, k_b, and v_b represent query, key, and value; x_f and x_b represent the forward and backward modalities (i.e., either r_i or d_i depending on the input order); *reshape* denotes dimensional transformation; $W_{qkv}()$ are linear projections used to generate queries, keys, and values; and α is a learnable hyperparameter.

Following cross-modal attention, wavelet decomposition was applied to split the frequency-enhanced features r_f and d_f into low-frequency components r_l (d_l) and high-frequency components r_h (d_h). The low-frequency components r_f and d_f were concatenated to form $f_{l,i}$. while the high-frequency components r_f and d_f were concatenated to form $f_{h,i}$. This dual-frequency processing preserved both global structure and fine-grained details: low-frequency components captured semantic-level information, while high-frequency components highlighted edges and textures, enabling comprehensive scene understanding. The formulation was as follows:

$$r_l, r_h = WD\left(r_f\right),\ d_l, d_h = WD\left(d_f\right) \tag{18}$$

$$f_{l,i} = Con\left(Cat\left(r_l, d_l\right)\right) \tag{19}$$

$$f_{h,i} = Con\left(Cat\left(r_h, d_h\right)\right) \tag{20}$$

where $WD()$ denotes the wavelet decomposition operation.

An adaptive gating mechanism was introduced to dynamically balance the contributions of different modalities, resulting in a fused representation w_{rd}. This mechanism allowed the network to modulate the relative importance of each modality based on the visual context of the scene,

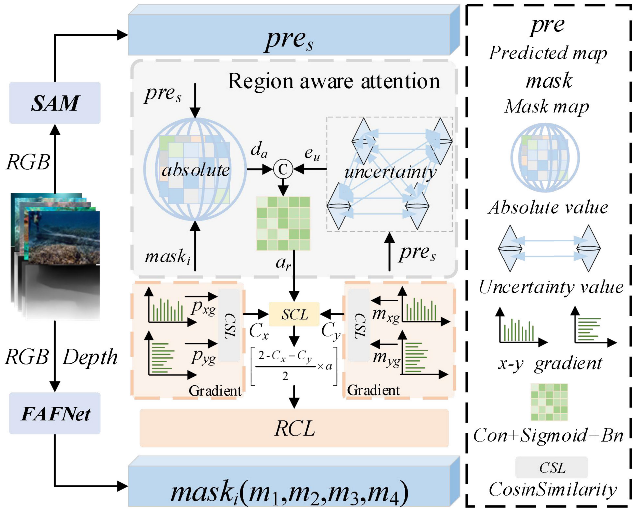

Fig. 4. Framework summary of RCL.

enhancing robustness under complex and variable underwater conditions. The formulation was

$$gate = Gate\left(Cat\left(r_{f,i}, d_{f,i}\right)\right) \tag{21}$$

$$w_{r,d} = r_{d,i} \times gate\,[0] + d_{r,i} \times gate\,[1]\,. \tag{22}$$

Finally, the low- and high-frequency components, $f_{l,i}$ and $f_{h,i}$, were concatenated with w_{rd} to produce the fusion feature, f_i

$$f_i = CBR\left(Cat\left(f_{l,i}, f_{h,i}, w_{rd}\right)\right) \tag{23}$$

where $CBR()$ denotes a sequential block consisting of convolution, batch normalization, and rectified linear unit activation.

The frequency cross-attention fusion module served as the core of the proposed MPG network, enabling effective multimodal feature integration by combining frequency-domain attention, cross-modal attention, wavelet decomposition, and adaptive gating.

D. Region-Aware Attention Collaborative Learning Loss

As indicated by the red arrow in Fig. 3, the RCL mechanism was applied during training. Fig. 4 illustrates the complete architecture of the RCL framework. RGB images were processed through SAM to generate predictions pre_s, while RGB and depth images were passed into the MPG to produce pseudo-mask $[m_1, m_2, m_3, m_4]$. Then, the SAM prediction pre_s was compared with each pseudo-mask from MPG to calculate the RCL. This comparison generated a collaborative learning loss that enabled cyclic knowledge transfer between SAM and MPG during training.

1) RCL: SAM-CLNet required an appropriate loss function to facilitate knowledge transfer from SAM (the teacher model) to MPG (the student network). Initially, we employed the Kullback–Leibler divergence (KLD) [53] to quantify the distributional difference between SAM and MPG outputs. However, SAM's strong prediction capability was not always effectively transferred to MPG, and in some instances, this transfer might impair certain performance metrics. This limitation stemmed from structural disparities and differing feature representation spaces between the networks, rendering basic KLD loss insufficient to capture intermodel knowledge gaps [54]. To address this, we introduced the RCL mechanism, which adaptively identified "difficult regions" and assigned them higher attention weights a_r by evaluating prediction discrepancy d_a and spatial uncertainty entropy e_u between SAM and MPG outputs. In addition, we incorporated a structural consistency loss [55], which preserved structural information by computing the cosine similarity of gradient directions [56].

We acknowledged that spatial regions contributed unequally to effective knowledge transfer. Locations showing large prediction differences between teacher and student models—and where the teacher model exhibited higher uncertainty—were defined as "difficult regions" that required increased focus during training. To this end, we designed a region-aware attention mechanism, the core operation of RCL, which emphasized areas with pronounced discrepancies between SAM and MPG predictions.

The region-aware attention mechanism first computed the absolute difference d_a between SAM and MPG outputs. Then, it calculated the spatial uncertainty entropy e_u of the SAM output. These two maps were concatenated along the channel dimension and passed through a series of convolutional layers followed by a sigmoid activation function to generate initial attention weights. Then, these weights were normalized to yield the final region-aware attention map a_r

$$d_a = |pre - m_i| \tag{24}$$

$$e_u = -\left[pre\log\left(pre+\varepsilon\right) + \left(1-pre\right)\log\left(1-pre+\varepsilon\right)\right] \tag{25}$$

$$a_r = BN\left(Sig\left(Con\left(Cat\left(d,e\right)\right)\right)\right) \tag{26}$$

where $BN()$ denotes the batch normalization operation.

After obtaining the attention map a_r, we applied KLD to measure the probability distribution difference between the teacher (SAM) and student (MPG) models. To emphasize learning from "difficult regions," the attention map a_r was directly integrated into the KLD loss formulation, enabling adaptive weighting of spatial discrepancies

$$L_{KLD_i} = KLD\left(pre, m_i\right) \times a_r. \tag{27}$$

The structural consistency loss, LSCLi, was computed in a similar manner. Specifically, we calculated the gradient maps p_{xg} and p_{yg} (for pre_s) and m_{xg} and m_{yg} (for m_i) in the x-direction and y-direction, respectively. Cosine similarity between the corresponding gradient pairs C_x between p_{xg} and m_{xg} and C_y between p_{yg} and m_{yg} was then evaluated to enforce structural and edge information transfer. The detailed formulations were

$$X_{xg} = GradientX\left(\,\right) \tag{28}$$

$$X_{yg} = GradientY\left(\,\right) \tag{29}$$

$$C_x = CosineSimilarity\left(p_{xg}, m_{xg}\right) \tag{30}$$

$$C_y = CosineSimilarity\left(p_{yg}, m_{yg}\right) \tag{31}$$

$$L_{SCL} = \frac{2 - C_x - C_y}{2} \times a_r \tag{32}$$

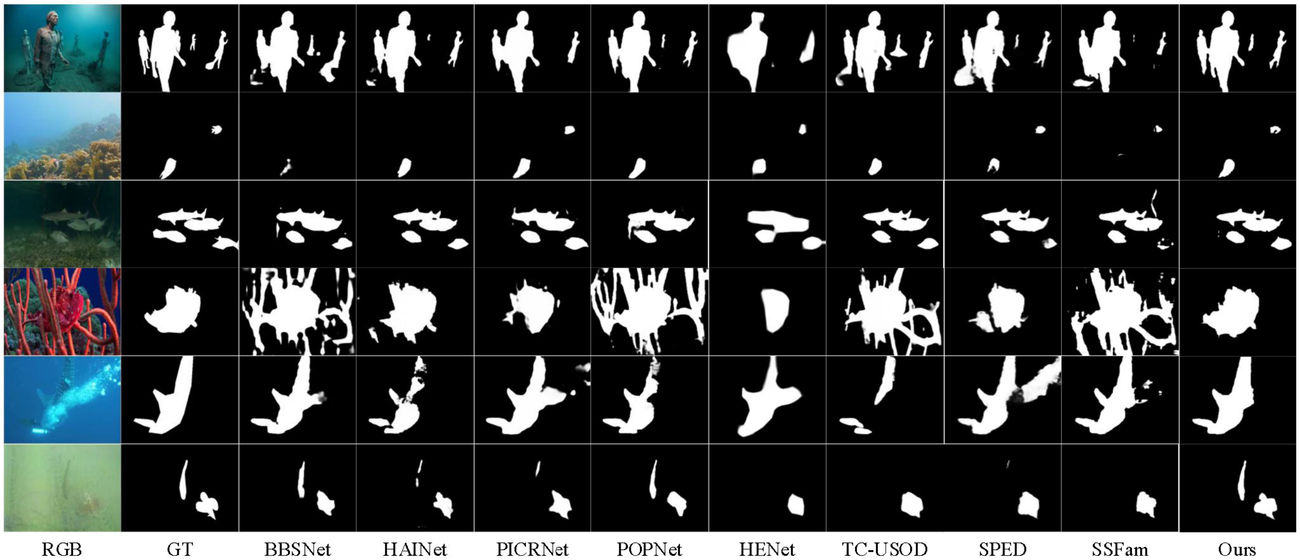

Fig. 5. Comparative visualization of different methods.

where X represents either pre_s and m_i, $GradientX()$ and $GradientY()$ compute the gradient values in the x- and y-directions, respectively, and $CosineSimilarity()$ denotes the cosine similarity loss function.

Finally, the individual losses L_{KLDi} and L_{SCLi} were summed to form the complete RCL loss L_{RCL}

$$L_{RCL} = \sum_{i=1}^{4} (L_{KLDi} + L_{SCLi}). \tag{33}$$

The RCL achieved its effect by identifying and emphasizing "difficult regions." It aligned the probability distributions of SAM and MPG using the KLD loss, while simultaneously preserving edge and structural consistency through the SCL. This dual-objective design enabled a closed-loop optimization mechanism, whereby SAM iteratively refined the quality of the masks generated by MPG and enhanced its own predictions.

E. Total Loss

The primary supervision losses for the SAM and MPG components combined binary cross-entropy loss and intersection-over-union loss [57], [58], applied to pre_s and m_i, respectively, using the corresponding GT labels

$$Loss_{SAM} = BI(pre, GT) \tag{34}$$

$$Loss_{MPG} = \sum_{i=1}^{4} BI(m_i, GT). \tag{35}$$

The final total loss used to train SAM-CLNET aggregated the SAM loss, the MPG loss, and the RCL loss

$$Loss_{total} = Loss_{SAM} + Loss_{MPG} + L_{RCL}. \tag{36}$$

IV. Experiments and Results

A. Datasets

We evaluated the performance of SAM-CLNet using the USOD10K [4] and USOD [3] datasets. Following the benchmark protocol established by Hong et al. [4], both variants of our model were initially trained on the USOD10K training set, which included 7178 samples. Performance was subsequently assessed on the USOD10K test set and the USOD dataset. The USOD10K dataset contained 10 255 images in total, divided into 7178 training samples, 2051 validation samples, and 1026 test samples. It spanned 70 salient object categories across 12 underwater scenes. The USOD dataset included only RGB images; we generated the corresponding depth maps using the same procedure applied to USOD10K. This dataset was used exclusively for model testing.

B. Evaluation Metrics

We assessed model performance using four widely adopted saliency detection metrics: S-measure (Sm), maximum E-measure (maxE), maximum F-measure (maxF), and mean absolute error (MAE) [19], [59], [60]. Sm measured structural similarity between the predicted saliency map and the GT, with scores ranging from zero to one, where higher values indicated better alignment. maxE captured both local and global image similarity in a binarized context, also on a zero-to-one scale. maxF quantified the harmonic meaning of precision and recall, reflecting the model's ability to detect salient regions accurately. MAE evaluated the average pixel-wise absolute error between prediction and GT, where lower values reflected higher precision. These metrics collectively offered a comprehensive assessment of saliency model performance [61].

C. Implementation Details

The proposed model was implemented using the PyTorch framework, and all experiments were conducted on a workstation equipped with an NVIDIA TITAN Xp GPU. To ensure fair comparison across models, most training and testing images were uniformly resized to 256 × 256 pixels. For networks whose backbones were incompatible with this resolution, we retained the original image sizes specified in their respective

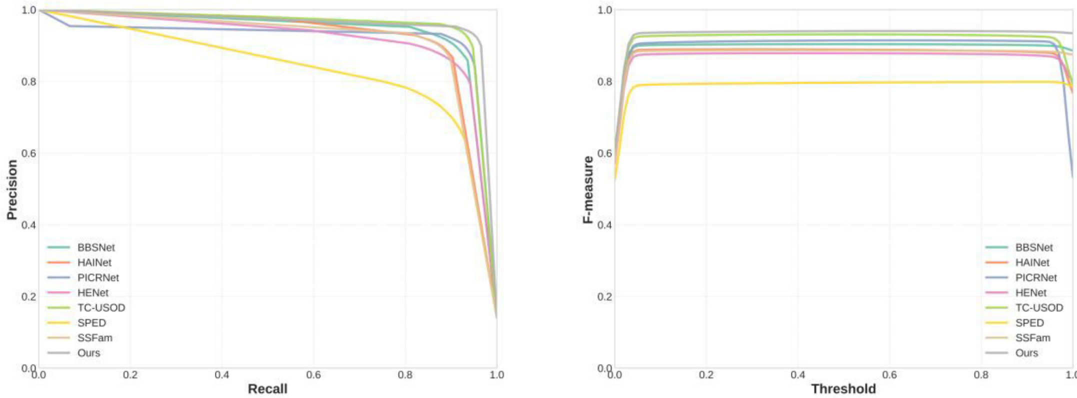

Fig. 6. Precision–recall curve (left). F-measure curve (right).

TABLE I
PERFORMANCE COMPARISON OF VARIOUS MODELS ON USOD DATASETS. RED AND BLUE TEXT INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

TNS-SOD	USOD10K				USOD			
	Sm↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
BBSNet$_{2020}$ [20]	0.8985	0.9516	0.8937	0.0274	0.8791	0.9246	0.8916	0.0509
TriTransNet$_{2021}$ [21]	0.7889	0.8479	0.7501	0.0659	0.8276	0.8936	0.8494	0.0768
HAINet$_{2021}$ [22]	0.8942	0.9402	0.8890	0.0314	0.8836	0.9247	0.8971	0.0504
CDINet$_{2021}$ [23]	0.7012	0.8567	0.7289	0.0896	0.6312	0.8178	0.7201	0.1489
PICRNet$_{2023}$ [24]	0.9175	0.9613	0.9181	0.0245	0.8923	0.9301	0.9079	0.0478
EGANet$_{2023}$ [25]	0.7789	0.8487	0.7324	0.0664	0.7746	0.8455	0.7751	0.0998
PopNet$_{2023}$ [26]	0.8974	0.9441	0.8920	0.0298	0.8854	0.9249	0.9013	0.0501
CATNet$_{2024}$ [27]	0.7174	0.7794	0.6395	0.0849	0.6858	0.7618	0.6605	0.1367
MAGNet$_{2024}$ [28]	0.7470	0.8295	0.6846	0.0729	0.7726	0.8443	0.7691	0.0958
HENet$_{2025}$ [29]	0.8708	0.9403	0.8521	0.0355	0.8587	0.9248	0.8701	0.0563
USOD	USOD10K				USOD			
SVAMNet$_{2022}$ [3]	0.7465	0.7649	0.6451	0.0915	0.8928	0.9315	0.8985	0.0548
TC-USOD$_{2023}$ [4]	0.9215	0.9683	0.9236	0.0201	0.8961	0.9319	0.9105	0.0455
SPDENet$_{2024}$ [5]	**0.9233**	**0.9688**	**0.9273**	0.0199	**0.9001**	0.9338	0.9134	**0.0436**
Dual-SAM$_{2024}$ [43]	0.9231	0.9681	0.9254	**0.0193**	0.8997	0.9347	0.9108	0.0438
HDANet$_{2025}$ [7]	0.9140	0.9608	0.9134	0.0235	0.8966	0.9357	0.9130	0.0442
STAMF$_{2025}$ [9]	0.9196	0.9646	0.9213	0.0207	0.8991	**0.9401**	**0.9178**	0.0421
SAM-SOD	USOD10K				USOD			
SAM$_{2023}$ [8]	0.8967	0.9411	0.8912	0.0290	0.8733	0.9122	0.8895	0.0560
SAM-Adaptor$_{2023}$ [12]	0.9238	0.9668	0.9229	0.0200	0.8974	0.9334	0.9132	0.0427
MDSAM$_{2024}$ [13]	0.8962	0.9456	0.8899	0.0276	0.8863	0.9274	0.9003	0.0484
SSFam$_{2025}$ [17]	0.8874	0.9356	0.8755	0.0317	0.8874	0.9318	0.9040	0.0486
Ours	USOD10K				USOD			
SAM-CLNet	**0.9296**	**0.9708**	**0.9296**	**0.0179**	**0.9026**	**0.9424**	**0.9204**	**0.0399**

publications. The training configuration was as follows: we used the AdamW optimizer with an initial learning rate of $1e^{-4}$ and a batch size of 4. Training was performed for 150 epochs, and the learning rate was reduced by a factor of 10 every 60 epochs.

A key strength of the proposed SAM-CLNet was its ability to directly process RGB–depth image pairs and generate saliency prediction maps in an end-to-end manner. This architecture required no additional preprocessing or postprocessing, thereby simplifying the workflow and enhancing computational efficiency [62], [63], [64].

D. Comparison With SOTA Methods

To evaluate the effectiveness of SAM-CLNet, we compared it with 20 SOTA SOD methods. This comparison included 10 RGB–depth SOD models originally developed for terrestrial natural scenes: BBSNet [20], TriTransNet [21], HAINet [22], CDINet [23], PICRNet [24], EGANet [25], PopNet [26], CATNet [27], MAGNet [28], and HENet [29]. In addition, we included six RGB–depth SOD methods designed for underwater environments: SVAMNet [3], TC-USOD [4], SPDENet [5], Dual-SAM [43], HDANet [7], and STAMF [9]. We evaluated SAM-CLNet against four recent SAM-based SOD models: the original SAM [8], SAM-Adapter [12], MDSAM [13], and SSFam [17]. To ensure a fair comparison, all models in the table were trained and evaluated using the same dataset and experimental setup as our proposed model. For the few methods without publicly available code, the results were taken directly from their respective publications.

As Table I shows, SAM-CLNet achieved superior performance across all evaluation metrics compared to the baseline methods, demonstrating its effectiveness in RGB–depth USOD. Fig. 5 provides visual comparisons across

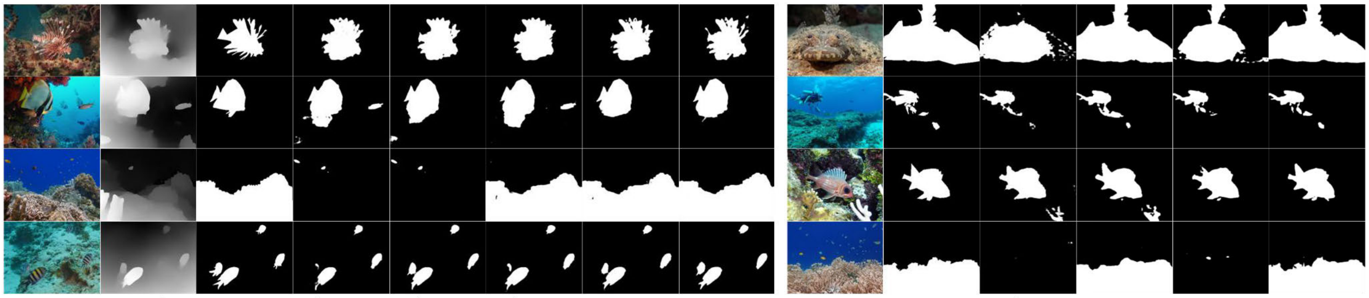

Fig. 7. MPG ablation visualization results (left). SAM-related component ablation visualization results (right).

TABLE II

MPG and SAM Ablation Experiments. Red and Blue Text Indicate the Best and Second-Best Results, Respectively. FCFA: Frequency-Cross Attention Fusion Module, FDA: Frequency Domain Attention, CMA: Cross Modal Attention, WD: Wavelet Decomposition, AG: Adaptive Gating

MPG(FCAF)					USOD10K				USOD			
RCL	*FDA*	*CMA*	*WD*	*AG*	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
					0.9150	0.9606	0.9121	0.0242	0.8862	0.9249	0.8996	0.0476
	√	√		√	0.9187	0.9632	0.9181	0.0219	0.8946	0.9312	0.9079	0.0436
		√	√	√	0.9151	0.9625	0.9136	0.0228	0.8937	0.9324	0.9096	0.0443
	√		√	√	0.9155	0.9629	0.9161	0.0217	0.8943	0.9329	0.9107	0.0439
	√	√	√		0.9157	0.9630	0.9157	0.0215	0.8951	0.9330	0.9113	0.0435
	√	√	√	√	0.9205	0.9637	0.9198	0.0205	0.8964	0.9335	0.9128	0.0431
√	√	√	√	√	0.9271	0.9692	0.9269	0.0187	0.9023	0.9419	0.9197	0.0404
SAM-CLNet					USOD10K				USOD			
MPG		*U-Adapter*			*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
					0.8770	0.9256	0.8599	0.0347	0.8766	0.9231	0.8928	0.0525
√					0.9175	0.9635	0.9147	0.0219	0.8906	0.9298	0.9041	0.0450
		√			0.9105	0.9550	0.9071	0.0245	0.8897	0.9269	0.9030	0.0474
√		√			0.9211	0.9653	0.9205	0.0200	0.8979	0.9345	0.9129	0.0426

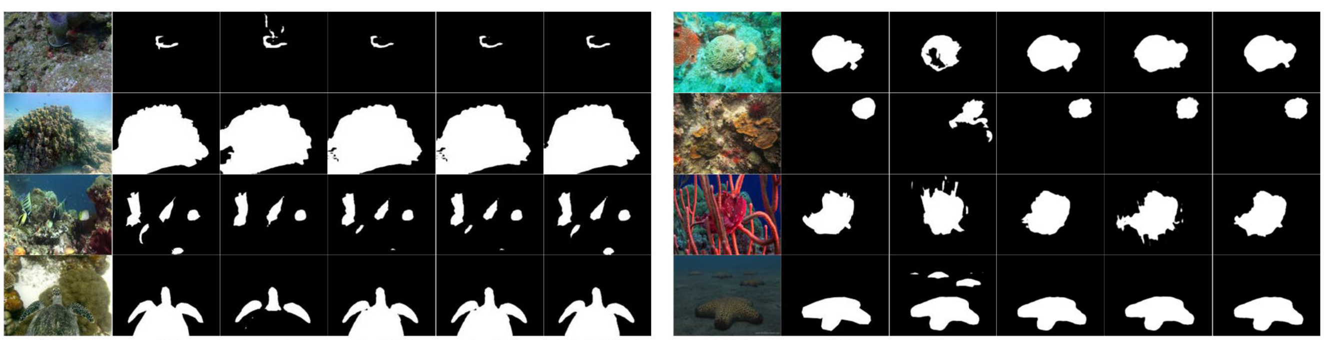

Fig. 8. RCL ablation experiment visualization results. SAM results (left). MPG results (right).

six representative underwater conditions: multiple objects, small objects, low lighting, multiple interfering objects, foreground–background similarity, and turbid water (corresponding to rows one through six). The 11th column shows the results of SAM-CLNet. Across all scenarios, SAM-CLNet consistently delivered robust and accurate predictions, even in challenging underwater environments. Fig. 6 shows the PR curve and the Fm curve.

E. Ablation Studies

To evaluate the contribution of each individual component within SAM-CLNet, we performed ablation studies on four key modules: MPG, SAM, RCL, and the U-Adapter. To isolate the roles of MPG and SAM further, we conducted additional ablation experiments excluding the RCL module.

1) As Table II shows, we evaluated the contribution of the frequency cross-attention fusion module in the MPG framework by selectively removing its four core components: frequency-domain attention, cross-modal attention, wavelet decomposition, and adaptive gating. A checkmark in the table indicated the inclusion of a component, while a blank cell indicated its exclusion. When all components were removed, RGB and depth features were fused via simple element-wise addition; when a single component was removed, alternatives

TABLE III

RCL Ablation Experiments. Red and Blue Text Indicate the Best and Second-Best Results, Respectively

SAM-CLNet			USOD10K				USOD			
Base	*SCL*	*KLD*	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
√			0.9211	0.9653	0.9205	0.0200	0.8979	0.9345	0.9129	0.0426
√	√		0.9277	0.9691	0.9278	0.0186	0.8994	0.9379	0.9164	0.0411
√		√	0.9252	0.9686	0.9241	0.0191	0.8992	0.9378	0.9148	0.0411
√	√	√	0.9296	0.9708	0.9296	0.0179	0.9026	0.9424	0.9204	0.0399
MPG			USOD10K				USOD			
√			0.9205	0.9637	0.9198	0.0205	0.8964	0.9335	0.9128	0.0431
√	√		0.9243	0.9684	0.9243	0.0193	0.8994	0.9375	0.9150	0.0415
√		√	0.9226	0.9657	0.9205	0.0200	0.8992	0.9373	0.9145	0.0416
√	√	√	0.9271	0.9692	0.9269	0.0187	0.9023	0.9419	0.9197	0.0404

TABLE IV

Absence of Region-Aware Attention Ablation in RCL. Red and Blue Text Indicate the Best and Second-Best Results, Respectively

SAM-CLNet			USOD10K				USOD			
Base	*SCL*	*KLD*	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
√			0.9211	0.9653	0.9205	0.0200	0.8979	0.9345	0.9129	0.0426
√	√		0.9247	0.9686	0.9249	0.0190	0.8968	0.9340	0.9116	0.0425
√		√	0.9242	0.9675	0.9233	0.0192	0.8984	0.9348	0.9121	0.0414
√	√	√	0.9269	0.9704	0.9278	0.0182	0.9005	0.9387	0.9153	0.0402
MPG			USOD10K				USOD			
√			0.9205	0.9637	0.9198	0.0205	0.8964	0.9335	0.9128	0.0431
√	√		0.9237	0.9683	0.9234	0.0195	0.8952	0.9329	0.9096	0.0433
√		√	0.9219	0.9665	0.9205	0.0200	0.8967	0.9340	0.9105	0.0422
√	√	√	0.9258	0.9688	0.9269	0.0206	0.9015	0.9355	0.9132	0.0430

TABLE V

SAM Adapter Comparative Experiments. Red and Blue Text Indicate the Best and Second-Best Results, Respectively

Models	USOD10K				USOD			
	Sm↑	*maxE*↑	*maxF*↑	*MAE*↓	*Sm*↑	*maxE*↑	*maxF*↑	*MAE*↓
SAM[8]	0.9175	0.9635	0.9147	0.0219	0.8906	0.9298	0.9041	0.0450
LoRA[44]	**0.9241**	**0.9682**	**0.9247**	0.0193	**0.9001**	**0.9359**	**0.9141**	**0.0418**
TransAdapter[66]	0.9224	0.9677	0.9228	**0.0192**	0.8989	0.9334	0.9139	0.0423
Dual-SAM[43]	0.9238	0.9666	0.9231	0.0199	0.8986	0.9348	0.9125	0.0427
SAM-Adapter[12]	0.9238	0.9668	0.9229	0.0200	0.8974	0.9334	0.9132	0.0427
MDSAM[13]	0.9229	0.9666	0.9219	0.0195	0.8974	0.9326	0.9115	0.0430
U-Adapter(Ours)	**0.9296**	**0.9708**	**0.9296**	**0.0179**	**0.9026**	**0.9424**	**0.9204**	**0.0399**

TABLE VI

Comparison of USOD Model Params and FLOPs

Model	*Params*(*M*)	*Flops*(*G*)
TC-USOD$_{2023}$[4]	125.80	29.60
SPDENet$_{2024}$[5]	115.66	41.46
STAMF$_{2025}$[6]	117.61	35.76
HDANet$_{2025}$[7]	159.50	80.80
SAM-CLNet	112.82	69.21

such as convolution or sigmoid were used. The experimental results demonstrated that MPG performed best when the full module was included, proving that the frequency cross-attention fusion module enhanced multiscale RGB-depth feature fusion. Finally, Table II contrasts MPG's performance before and after RCL-driven refinement; the results confirmed that SAM distilled knowledge into MPG via RCL, yielding substantial gains. Furthermore, Fig. 7 reports the ablation outcomes when RCL was removed.

2) In addition, Table II summarizes ablation results assessing the contribution of the SAM component. Performance declined markedly when both the MPG and the U-Adapter were removed, highlighting their essential role in supporting SAM. When the MPG-generated mask was used as a prompt input, performance improved, confirming that SAM was responsive to prompt cues. Notably, even in the absence of prompts, the inclusion of the U-Adapter led to measurable performance gains, emphasizing its use in enhancing feature extraction for underwater imagery. These findings were further illustrated in Fig. 7, where qualitative comparisons show that SAM struggles in downstream tasks, such as USOD10K, without appropriate prompt inputs. The visualizations also demonstrate that well-integrated adapters improve SAM's capacity to generalize underwater datasets. Together, these results underscored that both the MPG and U-Adapter were critical to enabling prompt-free adaptation of SAM in USOD.

3) As Table III shows, we conducted ablation experiments to evaluate the contribution of the RCL module. Specifically, we analyzed the individual and joint effects of

TABLE VII
PERFORMANCE COMPARISON OF VARIOUS MODELS ON THREE DATASETS. RED AND BLUE TEXT INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Models	NJU2K				NLPR				STERE			
	$S_m\uparrow$	$maxE\uparrow$	$maxF\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$maxE\uparrow$	$maxF\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$maxE\uparrow$	$maxF\uparrow$	$MAE\downarrow$
BBSNet_{2020} [20]	0.921	0.949	0.920	0.035	0.930	0.961	0.918	0.023	0.908	0.942	0.903	0.041
$\text{TriTransNet}_{2021}$ [21]	0.920	0.925	0.919	0.030	0.928	0.960	0.909	0.020	0.908	0.927	0.893	0.033
HAINet_{2021} [22]	0.912	0.944	0.915	0.038	0.921	0.960	0.915	0.024	0.907	0.944	0.906	0.040
CDINet_{2021} [23]	0.919	-	0.921	0.035	0.927	-	0.916	0.024	0.906	-	0.903	0.041
PICRNet_{2023} [24]	**0.927**	-	0.931	0.029	0.935	-	0.928	0.019	0.921	-	0.920	0.031
EGANet_{2023} [25]	-	0.922	0.883	0.033	-	0.966	0.912	0.020	-	0.924	0.870	0.042
PopNet_{2023} [26]	0.924	0.952	0.936	0.030	0.932	0.956	0.917	0.019	0.917	0.947	0.924	0.033
CATNet_{2024} [27]	0.925	0.935	0.902	0.030	**0.939**	0.968	0.916	0.018	**0.925**	0.935	0.902	0.030
MAGNet_{2024} [28]	0.928	**0.962**	**0.935**	**0.027**	0.938	**0.969**	0.931	**0.017**	**0.922**	0.956	0.920	**0.029**
HENet_{2025} [29]	0.918	0.912	0.899	0.034	0.932	0.960	0.892	0.021	0.916	0.933	0.895	0.036
SSFam_{2025} [17]	0.925	0.955	0.929	0.030	0.935	0.967	**0.933**	0.018	0.919	**0.957**	**0.927**	0.031
ISMNet_{2025} [70]	0.919	0.927	0.916	0.030	0.935	0.965	0.914	**0.017**	0.916	0.931	0.903	0.031
SAM-CLNet(Ours)	**0.932**	**0.964**	**0.939**	**0.024**	**0.941**	**0.973**	**0.938**	**0.015**	**0.925**	**0.960**	**0.924**	**0.028**

TABLE VIII
PERFORMANCE COMPARISON OF VARIOUS MODELS ON TWO DATASETS. RED AND BLUE TEXT INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Models	USODP				TJUP-USOD			
	$Sm\uparrow$	$maxE\uparrow$	$maxF\uparrow$	$MAE\downarrow$	$Sm\uparrow$	$maxE\uparrow$	$maxF\uparrow$	$MAE\downarrow$
BBSNet_{2020} [20]	0.9138	0.9575	0.9157	0.0287	0.9608	0.9891	0.9476	0.0048
LDF_{2020} [62]	0.9140	0.9587	0.9185	0.0258	0.9564	0.9919	0.9482	0.0040
TriTrans_{2021} [21]	0.9115	0.9589	0.9125	0.0236	0.9122	0.9758	0.9316	0.0066
HAINet_{2021} [22]	0.9097	0.9506	0.9075	0.0308	0.9600	0.9920	0.9493	0.0041
CDINet_{2021} [23]	0.9083	0.9491	0.9027	0.0296	0.9597	0.9893	0.9467	0.0042
S2MA_{2022} [72]	0.8586	0.9158	0.8438	0.0575	0.9528	0.9864	0.9338	0.0050
TC-USOD_{2023} [5]	0.9109	0.9549	0.9063	0.0240	0.9523	0.9877	0.9389	0.0043
CTIFNet_{2024} [69]	0.8981	0.9550	0.9004	0.0250	0.9043	0.9723	0.8968	0.0071
STAMF_{2025} [6]	0.9196	0.9646	0.9213	0.0207	**0.9630**	0.9926	0.9496	**0.0036**
SAM-CLNet(Ours)	**0.9274**	**0.9693**	**0.9291**	**0.0189**	**0.9623**	**0.9942**	**0.9510**	**0.0038**

its two loss components: symmetric KLD loss and SCL loss. The results indicated that both loss functions positively influenced model performance, with their combination yielding the highest accuracy and best overall saliency predictions. Qualitative comparisons are shown in Fig. 8, where the left and right panels display the outputs of the SAM and the MPG, respectively, under varying loss configurations. As shown in the first and second rows, applying either KLD or SCL alone improved knowledge transfer from SAM to MPG, enhancing object delineation relative to settings where RCL was excluded. These findings affirmed the central role of RCL in refining prediction consistency and strengthening cross-module collaboration in SAM-CLNet.

4) As Table IV shows, we further examined the impact of applying KLD and SCL losses to SAM-CLNet in the absence of region-aware attention. The results of the USOD10K dataset indicated that although these loss functions enhanced the training performance of both SAM and the MPG, their effects on the USOD test set were minimal or even detrimental. This suggested that simple knowledge alignment via KLD and SCL only narrowed output differences between the two models but did not adequately address cases in which MPG predictions diverged substantially from those of SAM. In contrast, the inclusion of a region-aware attention mechanism guided MPG to learn SAM's prediction behavior, particularly in complex underwater scenes, thereby improving the model's ability to generalize across datasets.

5) As Table V shows, we conducted comparative experiments using several established adapter methods, including LoRA [44], [65], TransAdapter [66], Dual-SAM [43], SAM-Adapter [12], and MDSAM [13]. Each of these five SAM-based adapters—originally designed for SOD tasks—was integrated into the SAM-CLNet framework for training. Experimental results demonstrated that the proposed U-Adapter outperformed the existing approaches, underscoring its superior adaptability and effectiveness for USOD.

6) As Table VI shows, we compared the parameters and FLOPs during the training phase of existing USOD models. We compared the parameters and FLOPs during the training phase of existing USOD models. Our SAM-CLNet, designed based on the large SAM model, maintains its training parameters and FLOPS within a relatively low range while achieving excellent performance.

F. Extended Experiment

To evaluate the effectiveness and generalization capacity of SAM-CLNet, we tested its performance on three RGB–depth SOD datasets—NJU2K, NLPR, and STERE—and two RGB–polarization USOD datasets—TJUP-USOD and

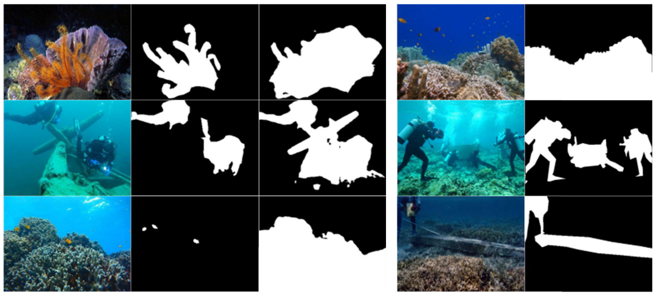

Fig. 9. Failed cases (left). Ambiguous photographs (right).

USODP. For RGB–depth experiments, we used 1485 image pairs from NJU2K and 700 pairs from NLPR for training, with the remaining samples used for testing. In the RGB-polarization experiments, 5692 images from TJUP-USOD were used for training and 1344 for testing. For USODP, we followed the same train–test split as USOD10K, replacing the depth modality with polarization data. As Tables VII and VIII show, performance scores were collected from original publications and other peer-reviewed sources [6], [47], [56], [67], [71], [72]. SAM-CLNet achieved strong results across multiple evaluation metrics on both RGB–depth and RGB-polarization datasets. These findings demonstrated that the proposed model maintained high effectiveness across varied input modalities and environmental conditions, further validating its generalization capability in diverse USOD scenarios.

G. Failed Cases and Analysis

Failure cases are illustrated in the left column of Fig. 9, while the right column presents ambiguous examples from the dataset. Although SAM-CLNet performs robustly overall, it exhibited occasional detection errors, especially in scenes containing multiple potential salient objects. For example, in the first row of the left column, both algae and coral appear visually prominent; however, the GT label included only algae, whereas SAM-CLNet identified both. Similarly, in the third row, the label included only fish, though coral was also clearly visible. On the right, the first-row image was labeled as coral in spite of the strong presence of a fish. These labeling inconsistencies in the USOD10K dataset introduced ambiguity, causing the model to learn conflicting saliency cues. Consequently, SAM-CLNet might identify multiple salient regions even when the GT included only one target, leading to apparent prediction errors that were, in part, dataset-induced.

V. Conclusion

We presented SAM-CLNet, a novel framework that addressed key challenges in applying the SAM to USOD. The architecture comprised three core components—SAM, an MPG, and RCL—which together formed a mutually reinforcing cyclical mechanism that enhanced detection in complex underwater environments. To preserve the original parameters of SAM, the proposed U-Adapter improved its adaptability by enabling dynamic prompt generation and facilitating feature fusion. The frequency cross-attention fusion module within MPG integrated RGB and depth information by combining frequency-domain attention, cross-modal attention, wavelet decomposition, and adaptive gating. RCL functions as the central connector in the framework, identifying difficult regions and enabling precise knowledge transfer from SAM's high-quality predictions to MPG. Extensive experiments demonstrated that SAM-CLNet achieved SOTA performance on the USOD10K and USOD datasets, outperforming existing methods across four key evaluation metrics. Additional evaluations further confirmed the framework's robustness and generalization across RGB–depth SOD and RGB–polarization USOD datasets. Although SAM-CLNet exhibited minor detection errors in isolated cases, it overcame the primary challenges of USOD. In future work, we aim to address performance limitations stemming from labeling inconsistencies in the datasets. Overall, SAM-CLNet provided a promising direction for advancing USOD research and established a practical transfer learning paradigm for adapting SAM to domain-specific downstream tasks. Regarding scalability and practical applicability, SAM-CLNet demonstrates several advantages for real-world deployment. First, the prompt-free design eliminates the dependency on manual annotations, making it practical for autonomous underwater vehicles operating in dynamic marine environments where human supervision is unavailable. Second, the modular architecture enables flexible adaptation to diverse underwater sensing platforms and environmental conditions without requiring complete model retraining. Third, extensive cross-dataset validation confirms the model's robustness under varied realistic conditions. For deployment on resource-constrained embedded systems, future work will explore knowledge distillation to transfer the superior performance of SAM-CLNet into lightweight networks while maintaining detection accuracy, thereby enabling real-time processing in time-critical underwater applications.

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